Speaker

DO | RASPBERRY PI | OUTPUT | SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses



Door chime

DETECT Button and chime setting → **DECIDE** Which chime should play? → **DO** Play the notes



Tiny fanfare machine

DETECT Count achievement-button presses → **DECIDE** Has the target been reached? → **DO** Play triumphant notes



Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import TonalBuzzer

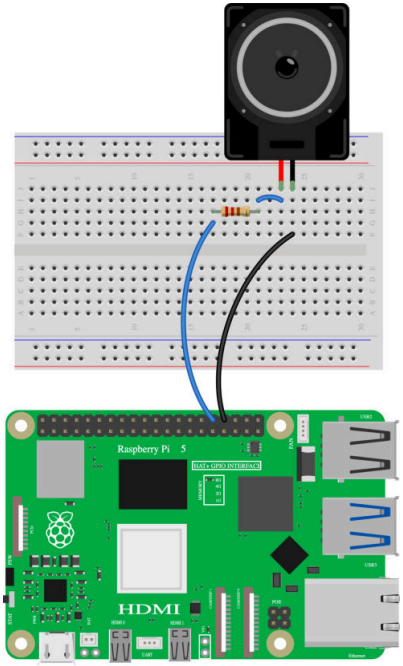
speaker = TonalBuzzer(12)

speaker.play("E4")
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

Wiring

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GPIO12 (pin 32) → 220 ohms → speaker red (+) lead; black lead → **GND** (pin 34)